

Namithaa V

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Education

Amrita Vishwa Vidyapeetham

CGPA: 8.34 | Aug 2023 – Present

B.Tech in Computer Science with specialization in Artificial Intelligence

Coimbatore, India

Coursework: Machine Learning, Deep Learning, Data Structures & Algorithms, Database Management Systems, Mathematics for Computing, Operating Systems, Probability, Natural Language Processing.

Projects

PCOSense – AI-driven PCOS Health Companion [\[GitHub: PCOSense\]](#)

- Collaborated on the design and developed a Flutter-Firebase mobile app for personalized PCOS care with ML-based cycle and fertility prediction, applying engineering principles for scalable, user-centric design.
- Implemented **real-time yoga pose tracking** using Google ML Kit for accurate posture detection.
- Developed an **AI-assisted Retrieval-Augmented Generation (RAG)** system with a **custom AI-powered web scraping pipeline** to collect **9000+ articles** and extract 4,500+ ingredients, storing structured data in a Neo4j knowledge graph for **on-demand retrieval** and nutritional insight generation.

Aval – Women Safety App [\[GitHub: AVAL\]](#)

- Developed a Women Safety App that enhances personal security through **real-time monitoring**, AI-driven analysis, and emergency assistance.
- Integrated GPS tracking, **geofencing**, **crime hotspot detection**, and **automated SOS alerts** to ensure immediate support in unsafe situations.
- Implemented **AI-based safety scoring with image analysis**, and **safer route suggestions** using real-time crime data.
- Added smart travel safety tips informed by regional crime patterns and built the platform using Flutter, Firebase, Google Maps API, Flask, and Twilio.

High-Fidelity Explainable AI Framework for Smart Grid Cyberattack Detection

- Proposed a **novel graph-language hybrid framework** for intrusion detection in smart grids, achieving **99.83% accuracy** on the DNP3 traffic dataset.
- Built a **ChebNet**-based Graph Neural Network (GNN) to learn behavioral patterns in DNP3 traffic via **k-NN graph** construction.
- Integrated **GNNExplainer** with Mistral-7B (fine-tuned via LoRA) to generate interpretable, structured threat reports from GNN-predicted anomalies.

Skills

Languages: Python, C++ , C , Java , HTML , CSS , SQL

Tools: GitHub, Android Studio, Hugging Face , Flask ,Tensorflow , Pytorch ,Neo4j ,ChromaDB

Achievements / Positions of Responsibility

- **Winner** – LaunchHacks IV (1,600+ participants): Secured 1st place for building PCOSense, an AI-powered mobile app for personalized PCOS management.
- **Patent Filed**– Designed and filed a patent for an IoT and ML-based Pollination Prediction and Optimization system for greenhouse farming.
- **Finalist** – Top 10 out of 500+ teams at the UNESCO Gender & Tech Hackathon for developing AVAL, a real-time women's safety mobile app.
- **Class Representative** – Serving as elected representative at Amrita Vishwa Vidyapeetham (2023 – Present), coordinating academic and administrative communication.
- **Solved 200+** algorithmic problems on LeetCode, strengthening data structures, algorithms, and problem-solving skills.